

JSS 2 MATHEMATICS – FIRST TERM

TOPIC 1: BASIC OPERATION OF INTEGERS / WHOLE NUMBERS I – STANDARD FORMS

1. Meaning of Standard Form

Standard form (scientific notation) is a way of writing very large or very small numbers as a product of a number between 1 and 10 and a power of 10. General form: $A \times 10^n$ where $1 \leq A < 10$ and n is an integer.

2. Whole Numbers in Standard Form

Example: $45,000 = 4.5 \times 10^4$

Example: $3,200,000 = 3.2 \times 10^6$

Example: $780 = 7.8 \times 10^2$

To convert: move the decimal point so that only one non-zero digit is to the left of it; the number of places moved is the power of 10 (positive if moved left).

3. Decimal Numbers in Standard Form

For numbers less than 1, the power of 10 is negative.

Example: $0.00045 = 4.5 \times 10^{-4}$

Example: $0.0078 = 7.8 \times 10^{-3}$

Move the decimal point to the right until the number is between 1 and 10; the number of places moved is the negative power.

4. Converting from Standard Form to Ordinary Form

If the power is positive, move the decimal point to the right. If negative, move to the left.

Example: $2.5 \times 10^3 = 2500$

Example: $6.1 \times 10^{-2} = 0.061$

5. Summary

Standard form writes numbers as $A \times 10^n$ with $1 \leq A < 10$. It is useful for very large and very small numbers. Practise converting both ways.

End of Topic 1